Airfoil Flow Prediction: *Where* and *Why* Mamba Blocks Enhance U-Net Architectures

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Problem: a fast surrogate for RANS

- CFD is *too slow* for design loops. A **U-Net** (DfpNet, 2.3 M params) maps free stream and airfoil mask (128×128) to p, v_x, v_y in one forward pass [4]. Pressure obeys a Poisson equation: every point depends on the whole domain.

$$\nabla^2 P = -\rho\, \partial_i \partial_j \left(U_i U_j + \overline{u_i' u_j'} \right)$$

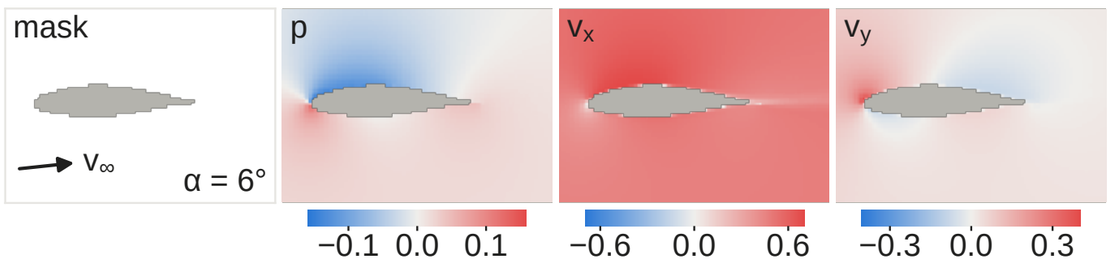


Figure 1: One validation sample: v_x is mostly free stream, v_y is induced by the airfoil.

Question: convs are local. Does a *global* block help, and *where*?
Expected: the largest gain on p.

Mamba instead of attention

- **Attention**: every pixel with every pixel, cost $O(L^2)$ in the L tokens [5].
- **Mamba**: a *selective state space model* scanned along the image in 4 directions (SS2D), cost $O(L)$ [1, 3].

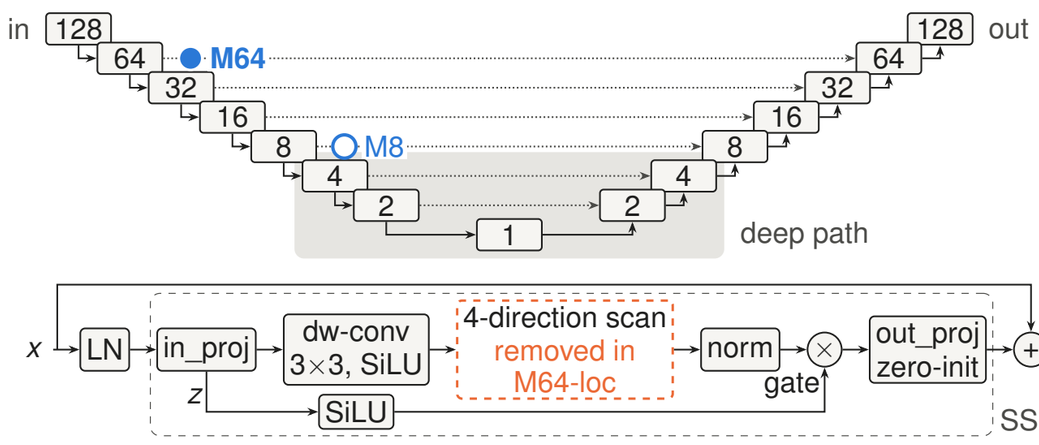


Figure 2: Top: DfpNet and insertion points. Bottom: residual Mamba (SS2D) block, zero-initialised: at the start the network *is* the baseline.

1. Deep levels: Mamba adds nothing, but can replace them

- A Mamba block at 8×8: no test gain (-0.01 [-0.34, $+0.26$]); it fits training data better, not new shapes.
- **Why**: the **deep path** below 8×8 (79% of the weights) already has a *global receptive field*.

	Deep path	Test error (5 seeds)
U-Net	convolutions down to 1×1	2.90%
Truncated	removed	3.47%
Mamba	**11 Mamba blocks at 8×8**	**2.79%**

Replacing the deep path with Mamba blocks *lowers the error* by 0.69 points *vs.* Truncated (95% CI = [0.26, 1.10]), and matches the U-Net.

2. At 64×64: v_y error -18% on the test set

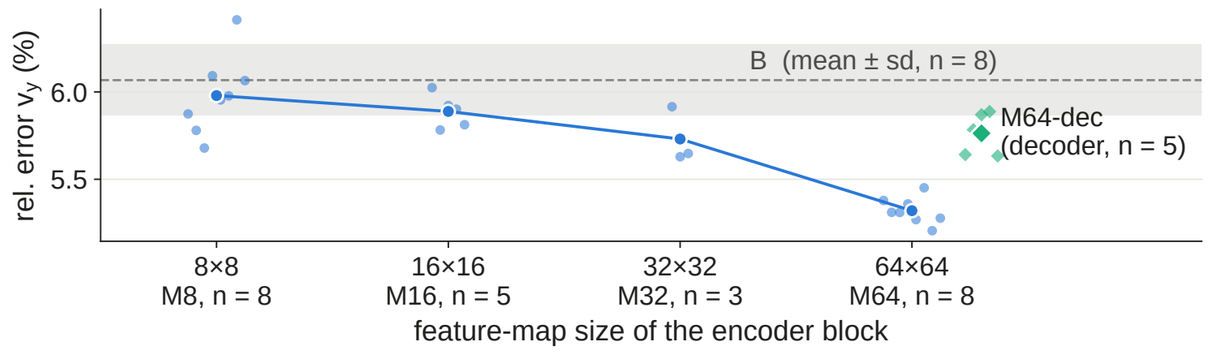


Figure 3: Validation v_y error *vs.* encoder block resolution (M8 – M64). Dots: seeds; large dot: mean; grey band: baseline B, mean ± s.d. The same block in the decoder (M64-dec) helps less.

- **Why 64 × 64**: here the U-Net applies a single conv., so the field is barely processed → the gain grows with resolution.
- **Confirmed on unseen airfoils** (pre-registered, 30 unseen profiles, 3 flows per profile): -1.17 points on v_y error (95% CI = [-2.47, -0.43]).
- **No demonstrated difference** on p, v_x and combined error ($0 \in$ 95% CI).

3. Ablation: the gain does not need the SSM

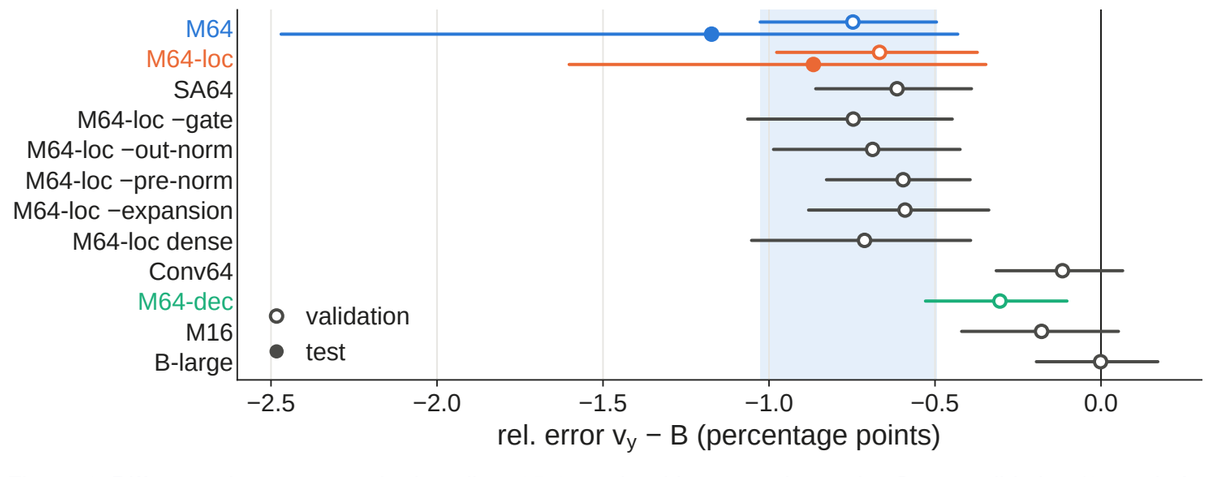


Figure 4: Difference in v_y error to the baseline, 95% two-level bootstrap intervals. Open: validation (8 seeds for M64, M64-loc, B-large; 5 for the others); filled: pre-registered test. Band: validation interval of M64.

- No scan (M64-loc): most of the gain stays. Attention (SA64) matches it, but trains 17× slower than B. A plain conv. block (Conv64) gives no proven gain.
- **Final architecture**: **U-Net + M64-loc** (+7 k parameters, 1.6× training time vs. B). Test: -0.87 [-1.60, -0.35], **confirmed**.
  Consistent with MambaOut [6].
- **Full 53.8k set** (validation, 3 seeds): M64-loc retains a smaller v_y gain (-0.22 points, -4.7% vs. B); wall-band v_y MAE is 6.1% lower.

4. Expected on p, found on v_y: why?

Our reading: consistent with the data, not tested directly.

- **The global part of p is already solved**: p is smooth, and a U-Net works like a multigrid V-cycle, the classic Poisson solver [2].
- **The p error sits in a tail of hard cases**: separated flows at high angle, missed by all models (2.5% of samples carry ∼15% of the error). Test: no demonstrated change in p error with M64 (mean -1.8 [-5.4, $+0.1$]).
- **The gain is local, at the wall**: within 4 px of the airfoil, v_y error drops by 11% (M64) and 10% without the scan (M64-loc) → v_y is almost entirely *airfoil-induced* (Fig. 1), so the gain shows on v_y but not on v_x.

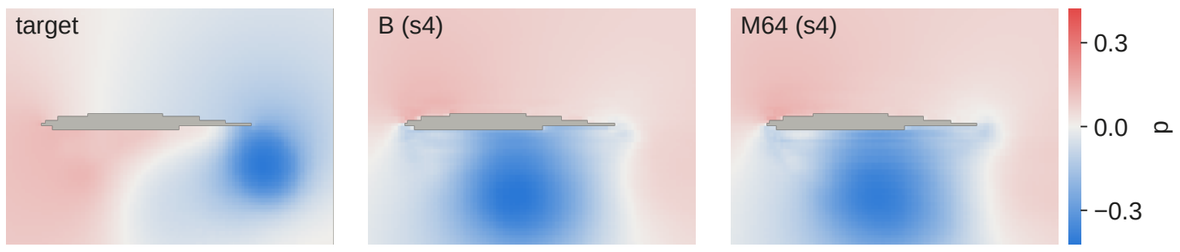


Figure 5: Pressure on a separated validation flow at $-22°$: the target sheds a vortex behind the airfoil; B and M64 keep it attached.

Take-home

- The U-Net already handles the global, elliptic part; it lacks processing at **high resolution**.
- There a **gated Mamba-style block** cuts the v_y error by 13 –18%; most of the gain remains **without the SSM**. With 8× more data, the gain roughly halves.

Project Repository

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